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Research article

Understanding different cultural ecosystem services: An exploration of rural landscape preferences based on geographic and social media data

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ABSTRACT

Rural landscapes offer a variety of cultural ecosystem services (CESs). However, the relationship between rural landscape characteristics and different CESs is still poorly understood. Therefore, this study explored the rural areas of Huzhou city, China, as a case study to assess the main rural landscape characteristics of different CESs based on public preferences. First, 148 scenic spots were classified into four CESs (physical, experiential, intellectual and inspirational), and the public preferences for each scenic spot were determined by combining tourists' scores obtained from social media and government assessment scores. Then, the landscape characteristic indicators were constructed from the natural, infrastructural and sensory perspectives by combining geographic and social media data. Finally, the random forest model was used to evaluate the public preferences for rural landscape characteristics overall and for different CESs. The word frequency analysis showed that, in addition to the nature landscape, infrastructure and service had a strong influence on public preferences. The relationship with rural landscape characteristics varied across different CESs. For physical CESs, the convenience of infrastructure played a greater role than natural landscape characteristics. Experiential CESs, on the other hand, were affected by natural landscape characteristics themselves. Intellectual CESs had higher requirements for both infrastructure and nature. Inspirational CESs included sensory evaluation indicators, in addition to their focus on natural landscape characteristics and infrastructure, indicating that this category of CESs was more concerned with inner experience. The use of social media data has enriched the dimensions of sensory elements and provided new ideas and information supplements for comprehensively understanding different CESs, thus better supporting the management, planning and protection of rural landscapes.

Credit author statement

Yongjun Li: conceptualization, data curation, methodology, software, writing-original draft, and writing-review & editing. **Lei Xie:** data collection and curation, software. **Ling Zhang:** data curation. **Lingyan Huang**: conceptualization, methodology. **Yue Lin:** writing-

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1. Introduction

With the advancement of urbanization and the improvement of people's living standards, human demands have gradually shifted from material needs to the pursuit of spiritual enjoyment and a yearning to get close to nature (Emborg and Gamborg, 2016; Guo et al., 2010). Rural areas have become an important tourism destination in contemporary society. Rural landscapes provide people with a wide range of cultural ecosystem services (CESs), such as recreation, science and education, cultural heritage and aesthetic functions (Van Zanten et al., 2014). Based on attention restoration theory (Kaplan et al., 1989), people's perceptions and sociocultural differences affect their preferences for different landscape characteristics and, ultimately, their access to cultural services (Gosal et al., 2019). Therefore, based on the perspective of the public preferences, studying the relationship between different CESs and rural landscape characteristics can lead to a better understanding of the formation of CESs and guide rural landscape enhancement strategies for strengthening the ecosystem service supply.

CESs refer to the nonmaterial benefits that people derive from ecosystems through spiritual enrichment, cognitive development, reflection, and aesthetic experiences and mainly include the following categories of CESs: recreation, aesthetics, science and education, cultural heritage and identity, spirituality and religion, and inspiration (Millennium Ecosystem Assessment, 2005). Of these, aesthetics is the most noteworthy CESs and has been studied extensively (Plieninger et al., 2013; Soy-Massoni et al., 2016; Zoderer et al., 2016). Schirpke et al. (2019, 2016) performed an in-depth study of the aesthetic dimension of mountain landscapes, mainly from the landscape pattern index, which showed that the public prefers landscapes with water-related and open visual fields. In terms of recreation, previous studies have focused more on the relationship between water bodies, trails, infrastructure and park and mountain recreation (Song et al., 2020; Tieskens et al., 2018). Among them, infrastructure was an important characteristic. In addition, the study by Blicharska et al. (2017) found that inspiration is seen as a benefit that can be combined with aesthetics or spirituality. Ecosystems also provide a rich source of inspiration for art, national symbols, folklore, architecture, and advertising (Millennium Ecosystem Assessment, 2005). However, studies related to inspiration have focused only on the value of inspiration for generating popular music and poetry (Coscieme, 2015; Dai et al., 2021). Cultural heritage is considered to be an expression of the relevant historical interactions between human and natural elements (Ortega Valcárcel, 1998) and plays an important role in promoting recreation and tourism development (Csurgó and Smith, 2021). However, most previous research has focused on the relationship between landscape characteristics and CESs in terms of recreation and aesthetics, with little consideration of the perspectives of science and education, cultural heritage and inspiration (Hølleland et al., 2017), possibly because recreational and aesthetic aspects are more easily measured in terms of economics and use (Hernández-Morcillo et al., 2013; Plieninger et al., 2013).

Previous studies of CESs primarily focused on urban areas, especially the public preferences for urban green spaces and parks (Krellenberg et al., 2021; Sinclair et al., 2020; Song et al., 2020; Teles da Mota and Pickering, 2021). Song et al. (2020) conducted an in-depth study on the landscape characteristics of urban parks from the perspectives of land use, park facilities, nearby infrastructure and residential population. The few studies of rural areas focused on a single land use type, mainly including mountains, forests, lakes, wetlands, agricultural lands, coastal dunes and protected areas (He et al., 2019; Ocelli Pinheiro et al., 2021; Schirpke et al., 2013, 2021a; You et al., 2022). Few studies have systematically considered all land use types and studied their relationship with rural landscape characteristics from the perspective of different CESs (Csurgó and Smith, 2021; Hegetschweiler et al., 2017).

Council of Europe (2000) considers that "landscapes are areas perceived by people, and landscape characteristics are the result of the action and interaction of natural and human factors". This means that the central component of landscape value is based on human perception (Ren, 2019). People mainly rely on sensory perception for spatial cognition, forming impressions of a space by observing its environmental characteristics, including the sound, light, and color in the landscape during various activities and judging whether they conform to their preferences. Therefore, the quantification of landscape characteristics is often used as an important way to study the spatial distribution of CESs, where natural and infrastructural elements are the basic landscape characteristics that distinguish different CESs (Manyani et al., 2021). Among the natural elements, land use types have the greatest influence on landscape characteristics (Schirpke et al., 2016; Wei et al., 2020). For example, indicators related to vegetation presence are considered one of the most important positive predictors (Coeterier J.F., 1996; Wang et al., 2016). In addition, diversity, patch density and total area have been found to be effective indicators of CESs (Jean-Christophe et al., 2020; Schirpke et al., 2013). Additionally, the accessibility and adequacy of infrastructure and amenities, such as restaurants, accommodations and transportation (Maaiah et al., 2021), have been shown to be important factors influencing human perception of the landscape (He et al., 2019; Khairabadi et al., 2020). However, people's perceptions of sounds, smells, and other sensory experiences in the landscape are not included in many studies, probably due to the difficulty of obtaining and quantifying these data; this should be addressed in future research (Han and Kim, 2010; Zoderer et al., 2016).

In recent years, the rise of geographic social media platforms (e.g., Flicker, Panoramio, and Instagram) has provided new ways to study preferences and landscape characteristics (You et al., 2022). Information such as geographic location, image content, and tags or keywords recorded with photos contained in social media data provide a wealth of knowledge about the spatial distribution of human-environment interactions (Ghermandi and Sinclair, 2019). The distribution density of shared photos and users' scores can be used to reflect the public preferences for different regional landscapes; it is more convenient and straightforward to compare these data than those obtained via traditional methods such as interviews and questionnaires (Grilli et al., 2021; Zhou et al., 2018). In addition, the relationship between public preferences and landscape characteristics can be explored through the content of tags and keywords, the landscape characteristics contained in the photos, or the geographical characteristics of the area where the punching point is located (Callau et al., 2019). Researchers have made attempts to mine semantic information (e.g., tags and keywords) (Gosal and Ziv, 2020), for example by using text mining techniques to classify terms and using the classification results as dummy variables to supplement prediction variables; however, the scarcity of tags or keywords and photo comments has limited the in-depth analysis of landscape preferences based on this approach (Jean-Christophe et al., 2020). Some current Chinese social media platforms, such as Ctrip, Qunar, and Dianping, can provide a large number of tourists' comments on landscape preferences. The social media reviews contain tourists' comments on the characteristics, strengths and weaknesses of attractions, as well as their subjective feelings (e.g., service quality, comfort, environmental conditions, etc.), which can provide valuable data for assessing the relationship between CESs and landscape characteristics (Zhang et al., 2016).

The aim of this study is to assess the dominant rural landscape characteristics of different CESs from physical, experiential, intellectual and inspirational aspects. Two key issues are addressed: (i) What are the dominant rural landscape characteristics that influence CESs? (ii) Are there differences in rural landscape characteristics across CESs? What are the dominant rural landscape characteristics across different CESs? To investigate these issues, a rural landscape characteristic indicator system based on geographic and social media data and word frequency analysis is proposed to assess public preferences from the natural, infrastructural and sensory perspectives. Then, random forest analysis is performed for different CESs to determine the relative importance ranks of the indicators.

2. Materials and methods

In this study, the landscape characteristics of the overall rural areas and different CESs were studied according to public preferences. For the sake of clarity, the analytical framework was divided into five steps (Fig. 1): (1) preprocessing Ctrip's big data, including determining scenic spots and analyzing the word frequency of comments; (2) identifying the different CESs (physical, experiential, intellectual and inspirational) of scenic spots according to word frequency analysis; (3) assessing people's preference scores by integrating public preference scores and government assessment scores; (4) determining the landscape characteristics indicators from the natural, infrastructural and sensory perspectives; and (5) evaluating the public preferences for rural landscape characteristics from the overall rural areas and different CESs and making planning recommendations.

2.1. Study area

This study focused on Huzhou city, Zhejiang Province, China. Huzhou covers an area of 5820 km^2 and is located along the coast of Lake Tai in the northern part of Zhejiang Province. Huzhou city is in the heart of the triangle created by the major mega-metropolitan areas of Hangzhou, Shanghai and Nanjing. According to the 2019 county survey, Huzhou had approximately 2.67 million registered permanent residents, the rural population was 1.49 million, and the number of domestic and foreign tourists reached 112.12 million in 2020. This county ranked among the top 30 best cities and was the 20th best tourist destination in China in 2018. The Zhejiang Provincial Government listed Huzhou as a pilot city for the special reform of rural tourism promotion and development in 2012.

Consistent with the national statistical data of China, the scope of rural areas in this study is divided according to the classification of the National Bureau of Statistics and the Regulations on Statistical Division of Urban and Rural Areas (Trial) (National Bureau of Statistics, 2009) (Fig. 2).

2.2. Data preprocessing

2.2.1. Social media data source and sample point retrieval

Ctrip is an important travel service app in China, with more than 135 million transaction users in 2018; it provides various services, including tickets to attractions, accommodation reservations, and transportation bookings (Zhang, 2019). The Ctrip website includes basic information about most travel products and their use and evaluation data, including tourist ratings, reviews, number of reviews, travel season and ticket prices for each scenic spot; these data can objectively reflect tourists' preferences and behaviors (Sun et al., 2020).

The scenic spot data were obtained from the Ctrip website using Python (Hou et al., 2019). The sample point retrieval procedure mainly included the following steps: the data were cleaned, and a coordinated conversion of the collected data was performed (CGCS 2000_3_Degree_GK_Zone_40); the scenic spots found in rural areas were selected, and the spots with fewer than 20 comments were removed; random and automatically generated comments were also removed to ensure the representativeness of the remaining comments. Finally, 148 scenic spots were selected from scenic locations in the countryside for inclusion in the analysis of rural landscape characteristics (Fig. 3b).

2.2.2. Word frequency analysis

In this study, the word segmentation module in Jieba, which is an open-access tool (https://github.com/fxsjy/jieba), was used to segment the comments on Ctrip into phrases. Jieba integrates two algorithms: string matching and statistical word segmentation (Huang et al., 2018). Jieba was used to scan a word graph based on a prefix dictionary to

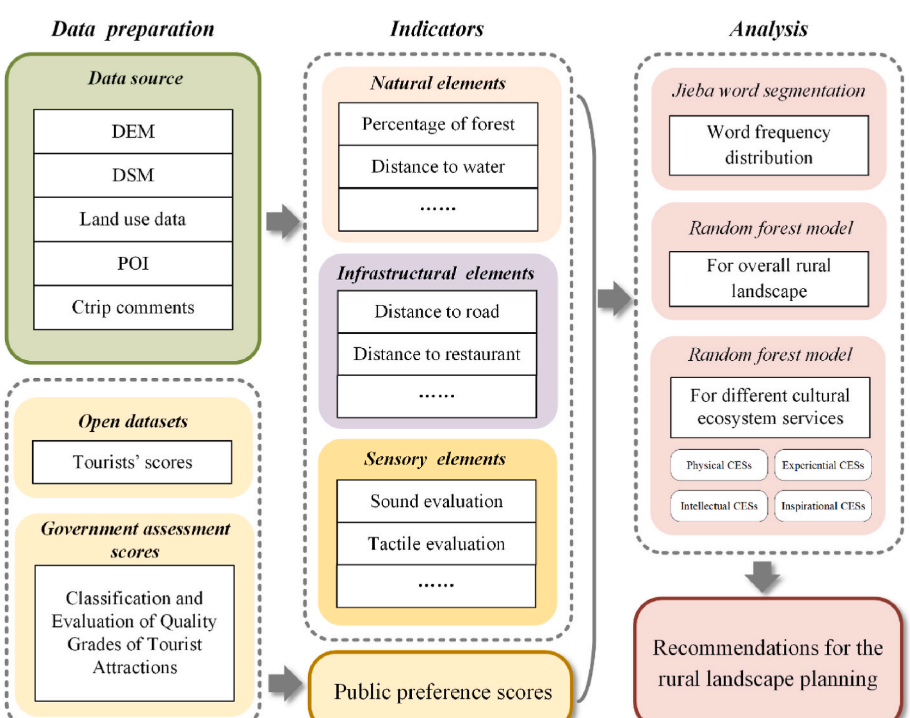

Fig. 1. Research framework.

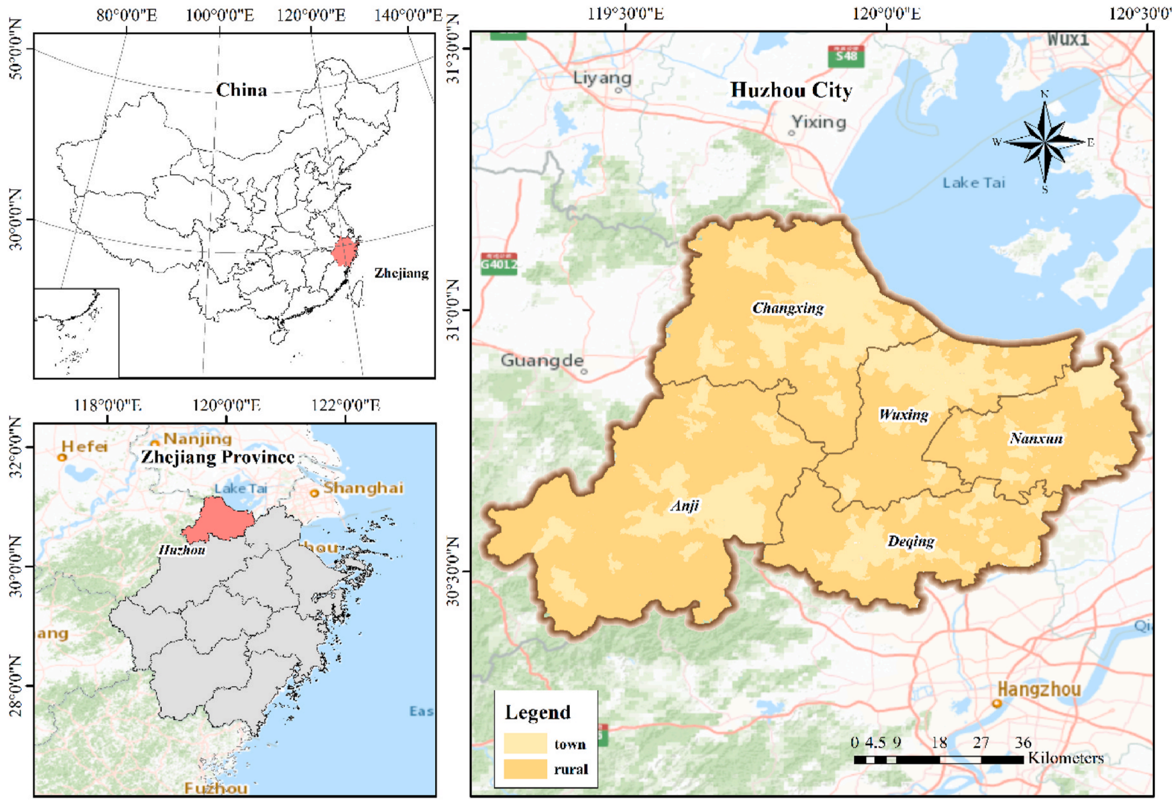

Fig. 2. Location of the study area.

generate a directed acyclic graph (DAG) containing all possible word formations. A dynamic programming method using corpus labeling was implemented to find the most likely path to obtain the largest segmentation combination based on word frequency. The keywords of the comments were extracted based on term frequency and inverse document frequency (TF-IDF), and repeated words such as onomatopoeia were eliminated. This study classified the extracted words into four major categories (biological and natural landscape elements, cultural landscape elements, perception elements and human elements) and their 20 subcategories (Table 1) (Komossa et al., 2020; Wang et al., 2021).

2.3. Identification of CESs

The common international classification of ecosystem services (CICES) was employed to classify CESs (Haines-Young and Marion Potschin, 2013). Accordingly, there were four categories of CESs: Physical (recreation); Experiential (aesthetics); Intellectual (scientific and educational, cultural heritage and identity); Inspirational (spiritual and religious, inspiration) (Clemente et al., 2019). Previous studies have shown that the terms associated with landscape characteristics are similar within the same landscape type; therefore, people's comments can effectively distinguish landscape types (Wartmann and Purves, 2018). The comment content of 148 scenic spots was analyzed by word frequency (Table 1). Different word frequencies represent different CESs characteristics (Table 2). The scenic spots were classified by CESs categories based on the results of the word frequency classification. Defined samples for each CESs were used to determine the accuracy and reliability of the classification. Then, the CESs were divided into four categories through k-means clustering analysis (Table 2). The k-means clustering algorithm is currently one of the most widely used clustering algorithms (Jain, 2010). The word frequency classification results were normalized by the z-score method (Zheng et al., 2018), and all the steps were executed in SPSS Statistics 25 (MacQueen, 1967).

To test the reliability of the word frequency classification procedure, Three professionals were invited to independently evaluate the content of 37,854 reviews of the 148 scenic spots to determine whether the k-means clustering results based on the word frequency results were related to cultural services. The agreement between word frequency classification and content-based classification was evaluated by the kappa coefficient (Cohen, 1960).

2.4. Assessing public preferences

To represent public preferences for rural landscapes, this study developed public preferences that integrated tourists' scores and government assessment scores (Fig. 3b). Tourists' scores of scenic spots were extracted from the Ctrip platform, including the comprehensive assessment results conducted by a mass of tourists. The government assessment scores were estimated using the Classification and Evaluation of Quality Grades of Tourist Attractions (Ministry of culture and tourism of the people's republic of China, 2012). According to this classification system, the evaluation criteria of national 5 A scenic spots include eight aspects: tourism transportation, tourism, resource environmental protection, sanitation, comprehensive management, post and telecommunications services, comprehensive management, and tourism shopping. These criteria provided a more objective and comprehensive evaluation of public preferences. The scores from the government evaluation and public participation were preprocessed and were both converted into a five-point scale. The formula is as follows:

$$S = S_{people} + S_{government} \tag{1}$$

where S_{people} and $S_{government}$ are based on a five-point scale and have the same calculation weight. $S_{government}$ is divided into five levels by the government (1 A = 1, 2 A = 2 … 5 A = 5). S represents the public preferences of the scenic spots.

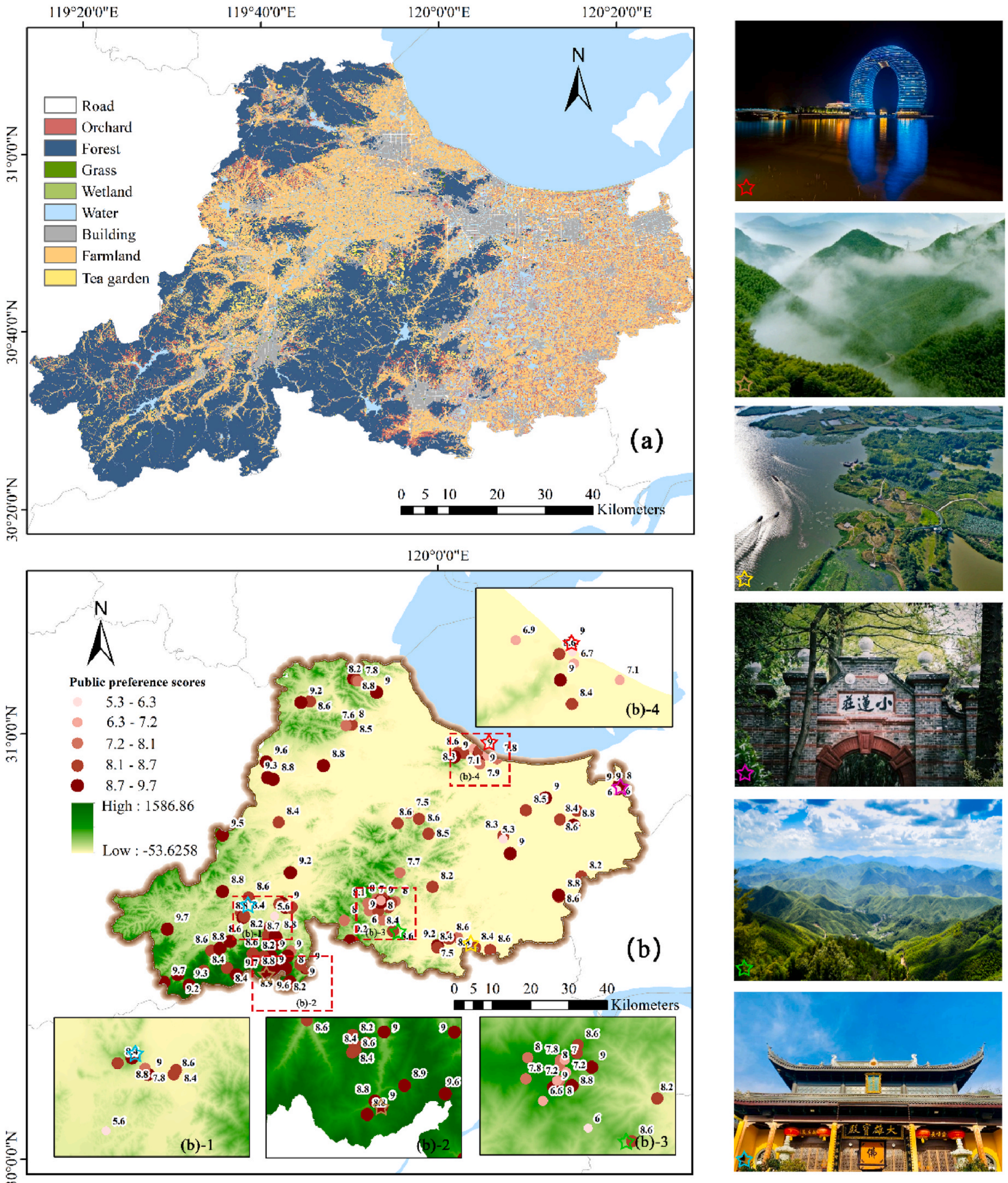

Fig. 3. The locations of scenic spots and public preference scores.

2.5. Landscape characteristics indicators

To assess public preferences for each scenic spot, a set of indicators were collected that represented the most important landscape characteristics mentioned in the literature (He et al., 2019; Schirpke et al., 2016; Tieskens et al., 2018), including natural elements, infrastructural elements and sensory elements (Table 3). A detailed description of all indicators can be found in Appendix A1.

2.5.1. Natural elements

The natural element indicators were mainly divided into three categories: landscape type composition, landscape metrics and air quality, which will be described in detail. Distance to water (D-water) mainly refers to the distance between the scenic spot and the nearest water, which was analyzed by Euclidean distance and nearest-neighbor analysis (He et al., 2019; Lin et al., 2022). Ctrip-animals were measured by word frequency analysis, which referred to the frequency of animal-related words in the total word frequency of each scenic spot (Section 2.2.2).

(1) Landscape type composition

The landscape type composition was calculated by the coverage rate of the landscape type within the viewshed of each scenic spot (Schirpke et al., 2021b). The landscape types were categorized into 9 groups: orchards, forest, grasslands, wetlands, farmland, tea gardens, roads, water

Table 1
Components of landscape characteristics.

Category	Sub-category	Word Examples
Biological and natural landscape elements	Plant type	Bamboo, flower, tea, peach blossom, ginkgo etc.
	Land type	Water, mountains, wetland, grassland, lake etc.
	Animal type	Chicken, bird, crocodile, giraffe, panda etc.
Cultural landscape elements	Building	Ancient building, former residence, ancient town, museum, parking lot, bridge, observation deck etc.
	Restaurant	Eat, breakfast, delicious, meat, snack etc.
	Homestay	Check in, big bed, resort, vacation etc.
	Tickets	Buy tickets, check-in, booking, package etc.
	Price	Cost performance, expensive, affordable, cheap, free etc.
	Entertainment	Climbing rock, taking pictures, climbing mountain etc.
	Distance	Far, near, high, low, slope, altitude etc.
Perception elements	Sound	Quiet, noisy, peaceful etc.
	Smell	Air, fresh, smelly, fragrant etc.
	Feel	Cool, heat, cold, temperature, feel etc.
	Vision	Pretty, vision, visibility, green, visit etc.
	Weather	Rain, rainbow, mist, snow, star etc.
Sense of place	Traffic	Self-driving, bus, traffic jam, mountain road, route etc.
	Service	Tourist guide, staff member, service attitude, management patient, recommended, great etc.
	Mood	Happy, relaxed, tired, cozy, disappointed, nervous etc.
	Time	Summer, holiday, weekend, summer vacation etc.
	People	Tourist, friend, children, family, old people etc.

Table 2
List of CESs.

CES types defined in this study (Clemente et al., 2019)	CES category (based on CICES)	Definition	Examples for classification
Physical	Recreation	Resources provided for recreational activities in the ecosystem (its biological and nonbiological elements).	Comments containing attractions for recreational activities (e.g., mountain climbing, skiing and rafting).
Experiential	Aesthetics	Feelings provided by the aesthetic characteristic of natural and seminatural landscapes and their biological and nonbiological elements	Comments containing descriptions of the landscape and beauty (e.g., mountain and river).
Intellectual	Scientific and educational	Research or educational activities conducted through the natural environment of the ecosystem and its biological and nonbiological components.	Comments containing attractions for educational training or research activities, (e.g., educational bases).
	Cultural heritage and identity	Value of the landscape, species or location to the local heritage and cultural heritage.	Comments containing cultural heritage or intangible cultural heritage (e.g., traditional buildings, local culture, cultural landscape and traditional practices).
Inspirational	Spiritual and religious	Landscapes, ecosystems and their elements that have religious or spiritual purposes.	Comments containing temples and religious attractions (e.g., churches, burning incense and worshiping Buddha).
	Inspiration	Landscapes, ecosystems and their elements used in art, architecture, advertising, local symbols, and folklore.	Comments containing attractions with art, publicity and local symbols (e.g., art gallery and music).

(including pit ponds, reservoirs, rivers, lakes), and villages based on the land use survey data in 2017 (Fig. 3a).

For the viewshed analysis, the digital surface model (DSM) was used as the observer's point of view for 148 scenic spots. The heights of ground objects were superimposed onto a digital elevation model (DEM) to generate a DSM. The average height in forest areas was 20 m (Wallentin et al., 2008), while that of shrubs was 2 m (Schirpke et al., 2013), and the average height of buildings was set to 10 m. The viewshed was analyzed in ArcGIS 10.2.

(2) Landscape metrics

The landscape metrics were analyzed by the visible range determined by the viewshed of each scenic spot, which was associated with the landscape type and was calculated by FRAGSTATS 3.3 (Huilei et al., 2017; Schirpke et al., 2013, 2019).

(3) Air quality

According to the Ambient Air Quality Standard (Ministry of Environment and Ecology, 2012), the number of days that each air quality monitoring site did not exceed the concentration limit was counted, defined by air quality index (AQI) < 100 and expressed it as a proportion of the number of days in a year. Ordinary kriging interpolation was then performed on these ratios (Lin et al., 2022).

2.5.2. *Infrastructural elements*

For the infrastructural elements, the location of the nearest road, town center, accommodation, restaurant, communal facility, and cultural attraction (e.g., museums, historic villages, and temples)were measured by Euclidean distance and nearest-neighbor analysis (Tieskens et al., 2018). Population density was reflected by the Weibo check-in data of each scenic spot, and the number check-ins was positively correlated with the population density.

2.5.3. *Sensory elements*

Most of the abovementioned indicators can be directly quantified to describe sociobiophysical characteristics, but some sensory indicators, such as smell quality and tactile quality, cannot be quantified and have rarely been considered in previous research. Therefore, according to the word segmentation results described in 2.2.2, the frequencies of positive words and negative words were calculated for the same word frequency types. The specific word frequency classification is shown in Table 1. The quality of this variable is expressed by subtracting the frequency of negative words from the frequency of positive words of the same word frequency type. The specific formula is:

$$Ctip\_Q_i = F_{positive} - F_{negative} \quad (2)$$

Table 3
Variables utilized for identifying rural landscape characteristics.

Dimension	Code	Meaning
Natural elements	NP	Number of patches
	TA	Total area
	PD	Patch density
	ED	Edge density
	CONTAG	Contagion index
	SHDI	Shannon's diversity index
	D-water	Distance to the nearest water bodies
	Air quality	Air Quality Index (AQI) compliance days
	Ctrip-animal	Animal word frequency in Ctrip
	P-water	Percentage of water in the viewshed
	P-farmland	Percentage of farmland in the viewshed
	P-orchard	Percentage of orchard in the viewshed
	P-forest	Percentage of forest in the viewshed
	P-wetland	Percentage of wetland in the viewshed
	P-grass	Percentage of grass in the viewshed
	P-tea garden	Percentage of tea garden in the viewshed
Infrastructural elements	D-road	Distance to the nearest road
	D-town center	Distance to the nearest town center
	D-accommodation	Distance to the nearest accommodation
	D-restaurant	Distance to the nearest restaurant
	D-communal facility	Distance to the nearest communal facility
	D-cultural attraction	Distance to the nearest cultural attraction
	population density	Weibo check-in data
Sensory elements	Ctrip-sound evaluation	Sound evaluation word frequency in Ctrip
	Ctrip-smell evaluation	Smell evaluation word frequency in Ctrip
	Ctrip-vision evaluation	Vision evaluation word frequency in Ctrip
	Ctrip-feel evaluation	Touch or feel evaluation word frequency in Ctrip

where $Ctip\_Q_i$ represents the evaluation quality of the i index at scenic spots Q, $F_{positive}$ is the positive word frequency and $F_{negative}$ is the negative word frequency. For example, the service quality of scenic spots can be evaluated by subtracting the word frequency associated with favorable service and negative service.

2.6. Statistical analysis

Random forest was adopted to explore the key indicators associated with the four CESs, and a random forest model was adopted for data mining. A random forest is a classifier consisting of multiple decision trees, which are grown by randomly selecting a subset of samples with replacements (Breiman, 2001; Huang et al., 2020). For random splitting Y_i of the variable groups that affect the rural landscape $X_1, X_2, X_3 \ldots X_j$, the public preference scores for Y_i are calculated as:

$$V(Y_i) = \sum_{n=1}^{j} p(Y_i = X_j)\left(1 - p(Y_i = X_j)\right) = 1 - \sum_{n=1}^{j} p(Y_i = X_j)^2 \tag{3}$$

where $V(Y_i)$ represents the public preference scores for Y_i and $p(Y_i = X_j)$ represents the probability of the estimated group. There are two important parameters in the model that need to be optimized: the number of spanning trees (Ntree) and the number of variables randomly selected at each node (Mtry). Ntree was set to the default value of 500, and Mtry was set to the square root of the number of input variables, as suggested by (Belgiu and Drăgu, 2016).

3. Results

3.1. Word frequency distribution

According to the word frequency classification results of each scenic spot, the scenic spots were divided into four categories through k-means clustering analysis. The results indicated that physical CESs had 47 scenic spots, experiential CESs had 49 scenic spots, intellectual CESs had 44 scenic spots and inspirational CESs had 8 scenic spots.

The relative word frequency at the scenic spots showed that building and service received considerable attention from tourists, with corresponding word frequencies reaching 0.92 and 0.71, respectively (Fig. 4a). Tourists were also affected by several elements, such as land, environment, distance, people and mood, for which the word frequency was between 0.2 and 0.3. The distributions of different word frequencies associated with the four CESs were basically similar, and the distributions were consistent with the overall word frequency distribution.

Tourists' preferred travel season was investigated based on the analysis of 39,493 comments on Ctrip (Fig. 4b). The results showed that 21.4% of tourists visited scenic spots in August, and there were also relatively high numbers of tourists in October and July, accounting for 13.5% and 11.2%, respectively. Winter months such as December and January represent the off-season for tourism. With respect to the four CESs, the travel times associated with different CESs were slightly different. The number of people who traveled to experimental attractions was the largest, accounting for 40.9% of tourists, while the number of people who traveled to inspirational attractions accounted for only 12.9%. Little difference was observed in the number of tourists visiting inspirational attractions throughout the year, while the number of visitors to other attractions was greatly affected by season. The peak tourism season for the physical, experimental and intellectual attractions was the same as the overall peak season, with visits mainly concentrated in August. Although the total frequency of experimental CESs was up to 0.409, the frequency of physical CESs and intellectual CESs varied relatively greatly with the seasons. The frequency of physical CESs and intellectual CESs varied greatly with the seasons, while the frequency of inspirational CES was relatively stable.

3.2. Overall rural landscape characteristics

The top 10 indicators that affected the views of the scenic spots were estimated by a random forest regression function (Fig. 5). Based on the public's preferences, the infrastructural elements and natural elements had an important impact on the overall rural landscape, and they had 6 and 4 indicators in the top ten, respectively. For natural elements, tourists were more concerned about the total area of people's viewshed, air quality, distance to water (D-water), and diversity of the scenic spots. Specifically, the positive indicator that greatly affected rural landscape preferences was the total area (TA). Air quality and D-water were relatively important indicators of the influence of natural elements. Shannon's diversity index (SHDI) is another natural element that affects aesthetic values. In terms of the infrastructural elements, D-road and distance to town center (D-town center) had pronounced effects on rural landscape preferences. People also paid more attention to cultural attractions, such as art galleries and museums. Restaurants, accommodations and communal facilities also played a central role in public preferences. In addition, the importance ranking results of all indicators of the overall rural landscape and different CESs were plotted in Appendix A2.

3.3. Rural landscape characteristics of different CESs

Fig. 6 shows the top 10 variables of the four CESs in terms of relative importance. The higher the relative importance value generated was, the stronger the influence of each variable on the CESs. The results showed that distance to accommodation (D-accommodation), distances

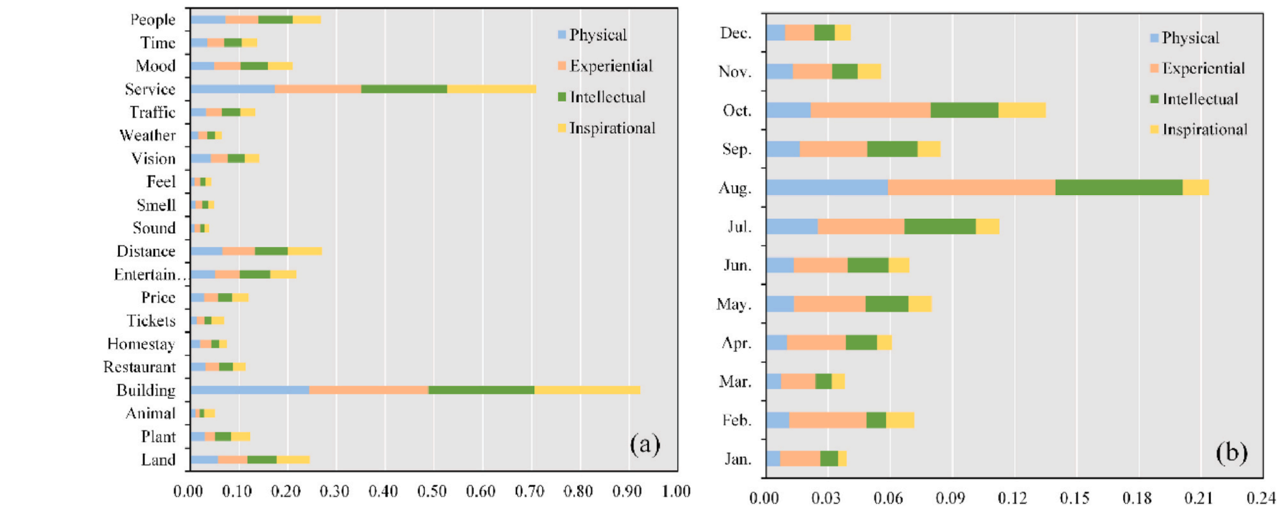

Fig. 4. (a) Relative word frequency of different keywords associated with the four CESs. (b) Relative frequency of tourism to the four CESs in different months.

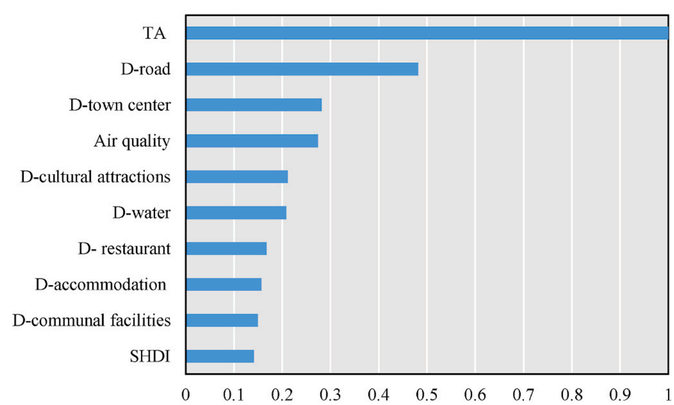

Fig. 5. The relative importance (standard-normalized) of the indicators of the overall rural landscape.

to restaurant (D-restaurant), air quality, TA, and distance to communal facility (D-communal facility) were indispensable, affecting three cultural services. For example, D-accommodation was among the top ten in the physical CES, intellectual CES and inspirational CES relative importance rankings.

The variables with the greatest relative importance were not the same for all CESs. D-restaurant played a central role in the physical CESs. The relative importance of two variables, D-accommodation and D-town center, followed closely, with values of more than 0.5. The main indicators all came from the infrastructural elements. The experiential CESs were mainly affected by natural elements. Patch density (PD) and TA were the two most important indicators, and the relative importance of the number of patches (NP) was also higher than 0.5. The experiential CESs were affected by the integrity and total area of the landscape. TA was vitally important in the intellectual CESs; the importance of D-road and edge density (ED) ranked second and third, respectively. In terms of inspirational CESs, D-communal facility, population density and air quality were the three most influential indicators. Moreover, sound evaluation and tactile evaluation were important in the CESs, indicating that tourists paid attention to subjective feelings.

4. Discussion

4.1. Important landscape characteristics based on word frequency analysis

In addition to natural landscape characteristics, supporting facilities such as parking lots, restaurants, and accommodations are also elements of greater interest to visitors. This is easy to understand because buildings provide people with the necessary places and facilities to carry out various activities. This result is consistent with the findings in the analysis of the relative importance of indicators for the four CESs, which found that different CESs showed a strong correlation with infrastructure. On the other hand, cultural landscape structures such as ancient villages and museums have profound historical and cultural value and are an important way for people to understand and experience history and cultural heritage. The CESs were also influenced by service quality. It is apparent that convenient transportation and service quality play important roles in tourist satisfaction (Moon and Han, 2018; Yin et al., 2021). Therefore, human factors, both hardware and soft services attached to the natural landscape, are both crucial from the point of view of forming CESs. Local climatic conditions, such as temperature and humidity, are also an important factors that affect CESs and can affect people's feelings about carrying out various activities (Steiger et al., 2020). For example, in the region where Huzhou is located, people like to go to the mountains in summer to escape from the heat and to engage in rafting activities. In addition, specific local climatic conditions are a prerequisite for some activities to be carried out. For example, in the mid and low latitudes, some mountainous areas retain snow fields for sports such as skiing due to the relatively lower temperatures in winter. Interestingly, inspirational CESs are less influenced by the seasons, which may be related to people's strong beliefs or to the fact that most of these venues are indoors.

4.2. Important landscape characteristics for different CESs

The importance of landscape characteristics varied with CESs. In terms of physical CESs, half of the ten indicators with the largest contributions were related to infrastructure. This indicates that in the formation of physical CESs, infrastructure plays a greater role than natural conditions (Tenerelli et al., 2016; Wolff et al., 2015). Good and convenient infrastructure not only provides the necessary conditions for relevant activities, such as mountain climbing, skiing and rafting, but is also important for people to be able to rest and recover after activities (Gidlow et al., 2019). In addition, feel evaluation also received greater attention, likely because the indicator related to physical sensation in

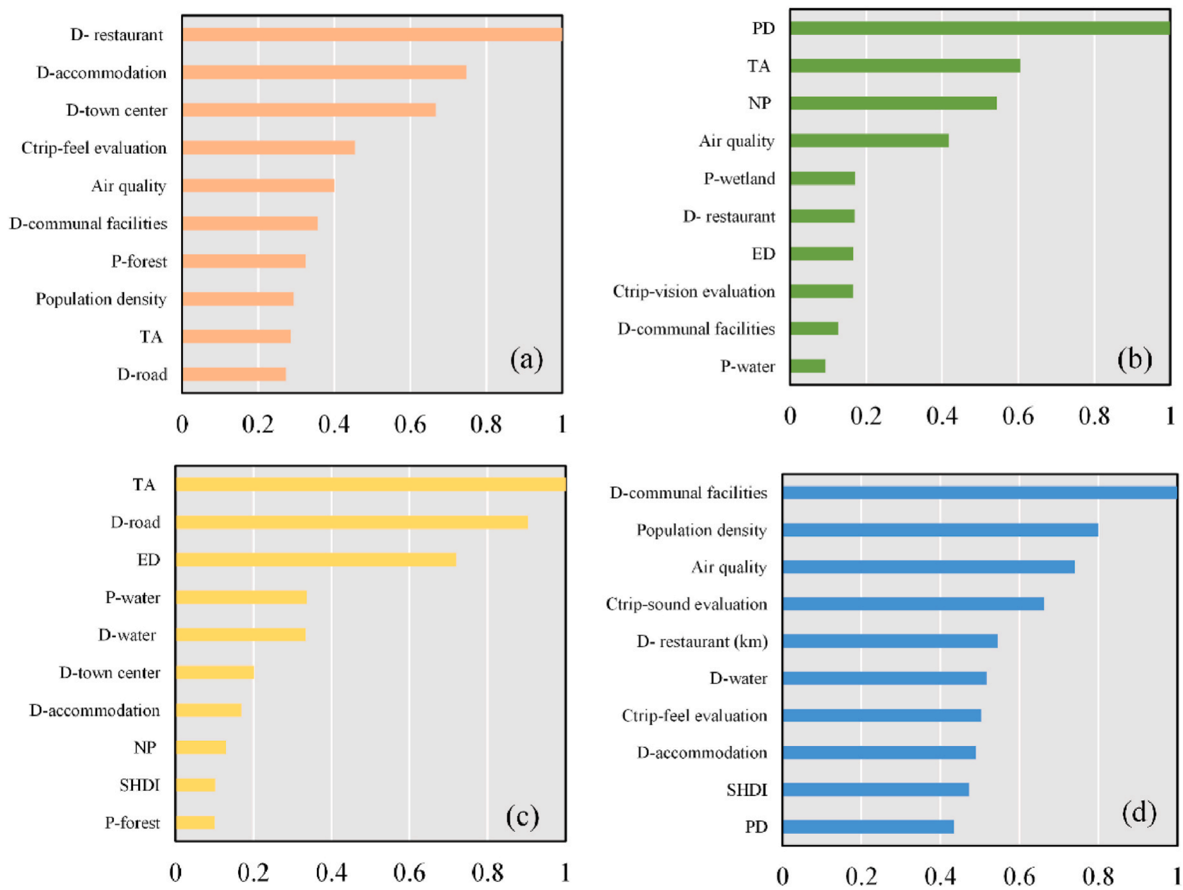

Fig. 6. The relative importance (standard-normalized) of the indicators of four CESs: (a) physical, (b) experiential, (c) intellectual and (d) inspirational.

this study was temperature, indicating that comfortable temperature was also an important consideration contributing to people's choice of leisure and recreational activities.

The importance of indicators of experiential CESs differed substantially from those of physical CESs. Eight of the ten most important indicators were those related to natural conditions. In contrast, among the facility-related indicators, only restaurant and common facilities were included. This indicated that for experiential CESs, the primary consideration was the qualities of the natural landscape itself, such as the continuity or diversity of the landscape, and the service facilities were secondary. Many studies have shown that people report a better feeling when they enjoy a continuous and magnificent natural landscape, such as continuous rice fields and the sea of flowers and forests (He et al., 2019).

For intellectual CESs, both natural and infrastructural elements were essential landscape characteristics, which are related to the diversity of science and cultural education. The government classifies scenic spots for scientific and cultural education into natural science museums (including nature museums, planetariums, zoos, botanical gardens, etc.), production sites, and national nature reserves (China Association for Science and Technology, 2022). Some scenic spots focused mainly on landscape indicators, such as landscape heterogeneity, landscape area, and distance to water. This may be related to the many science activities that focus on plant identification, farming experience, etc. In addition, the convenience of transportation was also observed to be an important indicator. This may be because group activities, such as visiting science museums, are influenced by accessibility.

Inspirational CESs can support the need for faith and the arts. Environments that provide inspiring CESs had lower population density and better air quality than other areas. Both touch and sound evaluation were important indicators. This is quite different from the performance of other CESs, implying that visitors to such attractions were more concerned with their inner feelings. Other studies have shown a strong relationship between the serenity of an attraction and the level of relaxation experienced (Watts and Pheasant, 2015).

4.3. Practical implications of social media data

Social media platforms can be a source of diverse data, including photo user days, user-generated tags and the content in the images, in addition to comment data, providing a scientific basis for revealing public landscape preferences and mapping the potential of aesthetic value (Maaiah et al., 2021; Schirpke et al., 2021a; Teles da Mota and Pickering, 2021). The applications of social media word frequency analysis were used to supplement sensory elements in this study. Landscape preferences were determined by a series of environmental attributes, including the overall sense of the landscape; the sense of its function, natural characteristics, spatial characteristics, and temporal development; the sense of place; and elements of perception (Coeterier J.F., 1996). Traditional GIS analysis methods can be applied to locate and quantify the characteristics of landscape elements, but they cannot determine attributes related to the social and cultural perceptions of landscapes. The present research centered around word frequency analysis of social media data to further understand people's dominant perceptions and determine the characteristic word frequency associated with landscapes. People's evaluations of scenic spots were obtained without creating restrictions in the form of questionnaires, thereby understanding public preferences and trends on sounds and smells. For example, in this study, preferences for religious science spots were closely related to observations regarding sound. Moreover, the proposed method was strongly generalizable and could be applied to many topics, such as green spaces, housing prices, scenic spots, and accommodations, to assess people's sensory evaluations (Su et al., 2021). For example, Wang et al. (2021) used word frequency analysis to explain park satisfaction and individual landscape characteristic satisfaction.

4.4. Study limitations

This study used geographic and social media data to compare the relationship between different CESs and landscape characteristics. This helps us to gain a deeper understanding of public preferences for different CESs and to apply this information for site selection and facility configuration when developing rural tourism. However, considering the influence of different cultural backgrounds on residents' preferences, the findings in this study may be applicable only in the Yangtze River Delta region of China. In addition, the adopted social media data do not include information on sociodemographic characteristics such as users' age, gender, and education level due to the issue of user privacy; therefore, potential differences in preferences for landscape characteristics among people with different backgrounds were unable to be assessed (Kalivoda et al., 2014; Sahraoui et al., 2016).

5. Conclusions

This study analyzed the relationship between rural landscape characteristics and CESs based on public preferences by combining geographic and social media data. The word frequency analysis of social media data showed that, in addition to the natural landscape, infrastructure and service quality also had a very strong influence on public preferences. Among the natural landscape characteristics, total area and distance to water were given a higher importance, indicating that people prefer continuous overall landscapes and water-friendly landscapes. There were differences in the relationship between different CESs and rural landscape characteristics. For physical CESs, the convenience of infrastructure, such as restaurants and accommodations, and transportation played a greater role than natural landscape characteristics due to the greater dependence on man-made facility conditions for the development of the various activities associated with physical CESs. Experiential CESs, on the other hand, were associated with the natural landscape characteristics themselves, especially the visual perceptual experience brought by a continuous landscape or a diversity of landscapes. Intellectual CESs include scientific and cultural education and had higher requirements for both infrastructure and nature. Inspirational CESs included sensory evaluation indicators, in addition to natural landscape characteristics and infrastructure, indicating that this category of CESs is driven by inner experiences. The main contribution of this study is to combine geographic and social media data research to provide a method to study the influence of rural landscape characteristics on CESs and to analyze the relationship between different CESs and rural landscape characteristics, which can guide decision making regarding landscape sustainability and visitor management (Angelstam et al., 2021; Fagerholm et al., 2021).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jenvman.2022.115487.

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